

CASE STUDY SUBMISSION

For: JLL

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Executive Assessment

Across the review of the draft charity-donor-outreach skill, 23 opportunities for improvement (fleshed out further in “Detailed Findings”) across three primary architectural failures:

Trust: The system generates unsupported factual claims, including fabricated staff names, inferred donor information, and unverified fundraising statements.

Determinism: Business rules, donor data, and arithmetic are embedded inside prompt instructions instead of deterministic software, preventing repeatable execution and auditability.

Operational Control: The workflow lacks validation, exception handling, human review, and performance monitoring, allowing data quality issues to propagate directly into donor communications.

These areas of failure introduce unnecessary operational risk and make consistent execution impossible as donor volume grows. As a function of these observations, what follows is an enumeration of A) proposed architectural principles, B) expected impact of these shifts, and the C) detailed findings that inform the proposed principles.

A. Architectural Principles

Separate Deterministic Logic from Language Generation: Calculations and rules belong in code (specifically the python script provided in the attached), which is exact and repeatable. The model’s responsibility must be limited to transforming verified inputs into donor-facing communication.

Establish a Single System of Record: Donor data must live in one governed system. Instruction files must never contain donor records, as maintaining duplicate records creates conflicting sources of truth and makes deterministic outputs impossible.

Ensure Every Claim Traces to an Input: A letter should only state what a verified input supports. Unavailable facts must be omitted rather than invented.

Manage Exceptions, Never Assumptions: Missing or inconsistent data should surface as visible exceptions for human resolution, never as silent guesses.

Implement Human Governance: Everything the system produces must be a draft, requiring approval by a named staff member before reaching a donor.

Measure Performance: A production workflow requires measurements to catch problems early and demonstrate value.

B. Expected Impact

The transition absorbing this approached to governed architecture produces four measurable improvements:

Factual Accuracy: By grounding the model in validated inputs, we eliminate the path for fabricated staff, donors, and statistics.

Repeatable Execution: Moving logic to deterministic software ensures that identical donor inputs always yield identical, audited outputs.

Operational Scalability: Modularizing the workflow allows the system to grow from 50 to 500,000 donors without requiring modifications to the skill logic.

Governance: Transitioning to a draft-and-approve workflow ensures accountability and professional oversight for all donor communications.

C. Detailed Findings

Key: (1) Skill Activation | (2) Workflow Steps | (3) Donor Tiers | (4) Campaign Types and Messaging | (5) Request Amount Calculation | (6) Salutation Rules | (7) Donor Giving Histories | (8) Letter Template | (9) Closing Instruction | (10) Controls Missing from the Skill Entirely

Trust

Matching-gift claims (4): The instructions permit promises of matching funds without confirmed sponsorship. Risk: Factual misrepresentation creates legal and reputational exposure. Recommendation: Restrict matching language to campaigns with a confirmed sponsor, ratio, and deadline.

Invented staff (3): The system is instructed to invent personal relationship manager names where none exist. Risk: Donors will contact the organization asking for non-existent staff, damaging trust. Recommendation: Map signer name and title to required campaign setup fields.

Unverified figures (4): The system fabricates event registration statistics. Risk: The assistant provides verifiable, false statistics in donor communications. Recommendation: Only cite registration figures if provided as a verified input.

Guessed demographics (6): The system infers titles and gender from names. Risk: Guessed titles often result in misgendering or stripping earned professional titles. Recommendation: Use provided data only; default to full names for major donors and first names for others.

Template blanks (8): The template includes placeholders with no defined data source. Risk: The model will invent plausible but incorrect information to fill the blanks. Recommendation: Map every template blank to a required field in the campaign setup.

Determinism

Arithmetic errors (5): The language model executes a multi-step formula to calculate request amounts. Risk: Language models are unreliable for repetitive arithmetic; error rates will spike as donor volume increases. Recommendation: Move all calculations into a deterministic program.

Ambiguous order of operations (5): The request formula lacks a clear order of operations. Risk: Ambiguous rules lead to inconsistent request amounts for similar donor profiles. Recommendation: Define the order explicitly: base, loyalty, volunteer, campaign multiplier, then round.

Flat request amounts (5): Bronze donors receive a flat \$150 request. Risk: A flat figure severs the request from the individual donor relationship. Recommendation: Base on % of donors largest gift + minimum (i.e. Base the request on 125 percent of the donor's largest gift, with a \$75 minimum)

Volunteer bonus logic (5): A flat \$100 volunteer addition is applied regardless of request size. Risk: For small requests, the addition creates a disproportionate, out-of-touch jump. Recommendation: Cap the addition at the lesser of \$100 or given % of amount (ie. 20 percent of the running amount).

Data duplication (2, 7): Donor information is held in both uploaded files and embedded in the skill. Risk: Conflicting sources of truth make repeatable execution impossible. Recommendation: Treat the uploaded file as the exclusive source of truth.

Tier/Status conflation (3): Value (Platinum/Bronze) and Recency (Lapsed) are merged into one category. Risk: Donors qualify for contradictory treatments. Recommendation: Track value and recency as independent variables.

Operational Controls

Broad activation criteria (1): The skill activates on virtually any mention of communication. Risk: Frequent false activations erode staff confidence. Recommendation: Activate only when a donor list is provided for fundraising outreach.

Scalability limitations (2): Chat output is unsuited for batch review. Risk: Process fails as volume grows. Recommendation: Generate discrete files and summary spreadsheets.

Missing campaign types (4): Unrecognized campaign types silently default to Annual Fund. Risk: Emergency appeals may be sent with standard messaging. Recommendation: Stop and request user selection if campaign type is missing.

Lack of reasonableness bounds (5): The formula has no minimum or maximum bounds. Risk: Bad data results in implausible requests. Recommendation: Apply a \$25 minimum and a ceiling of 150 percent of the largest gift.

Suppression handling (10): The workflow ignores deceased or do-not-contact flags. Risk: Contacting deceased donors is a major operational failure. Recommendation: Honor suppression fields prior to any generation.

Lack of human review (10): No review gate exists. Risk: Errors propagate directly to the donor. Recommendation: Route all output through a review queue.

Measurement gaps (10): The skill tracks no performance metrics. Risk: System degradation goes unnoticed. Recommendation: Track exception volumes and reviewer edit rates.

Assumption instructions (9): The system is told to assume values if fields are missing. Risk: Silent assumptions result in incorrect data delivered with confidence. Recommendation: Route incomplete records to an exceptions report.

Tier label contradictions (7): Donors are labeled with tiers that contradict their actual giving history. Risk: Revenue is left on the table due to misclassification. Recommendation: Recompute tiers from history and flag disagreements.

Personalization priority (2): No priority is set for personalization depth. Risk: Inconsistent personalization leads to model hallucination. Recommendation: Personalize in a fixed, verified order.

Lapsed donor tone (3): The system uses an apologetic, guilt-based tone for lapsed donors. Risk: Guilt-based framing depresses response rates. Recommendation: Use a warm, welcoming tone focused on past support.

Apologetic messaging (3): The system includes offers of tote bags and restart journeys. Risk: Unnecessary offers dilute the core call to action. Recommendation: Omit secondary offers and focus on a single renewed gift.